

2. This question is about acids and alkalis. Lisa started filling in the following table.

Chemical name	Chemical formula	Colour with Universal Indicator	pH	Acid, alkali or neutral
sulfuric acid	H_2SO_4	red	1	acid
hydrochloric acid	HCl	red		acid
calcium hydroxide	$\text{Ca}(\text{OH})_2$	purple	12	alkali
sodium hydroxide	NaOH		14	alkali
water		green	7	

(a) Complete the table.

[4]

(b) Lisa added magnesium to sulfuric acid. She recorded the temperature of the mixture during the reaction.

The word equation for this reaction is shown below.

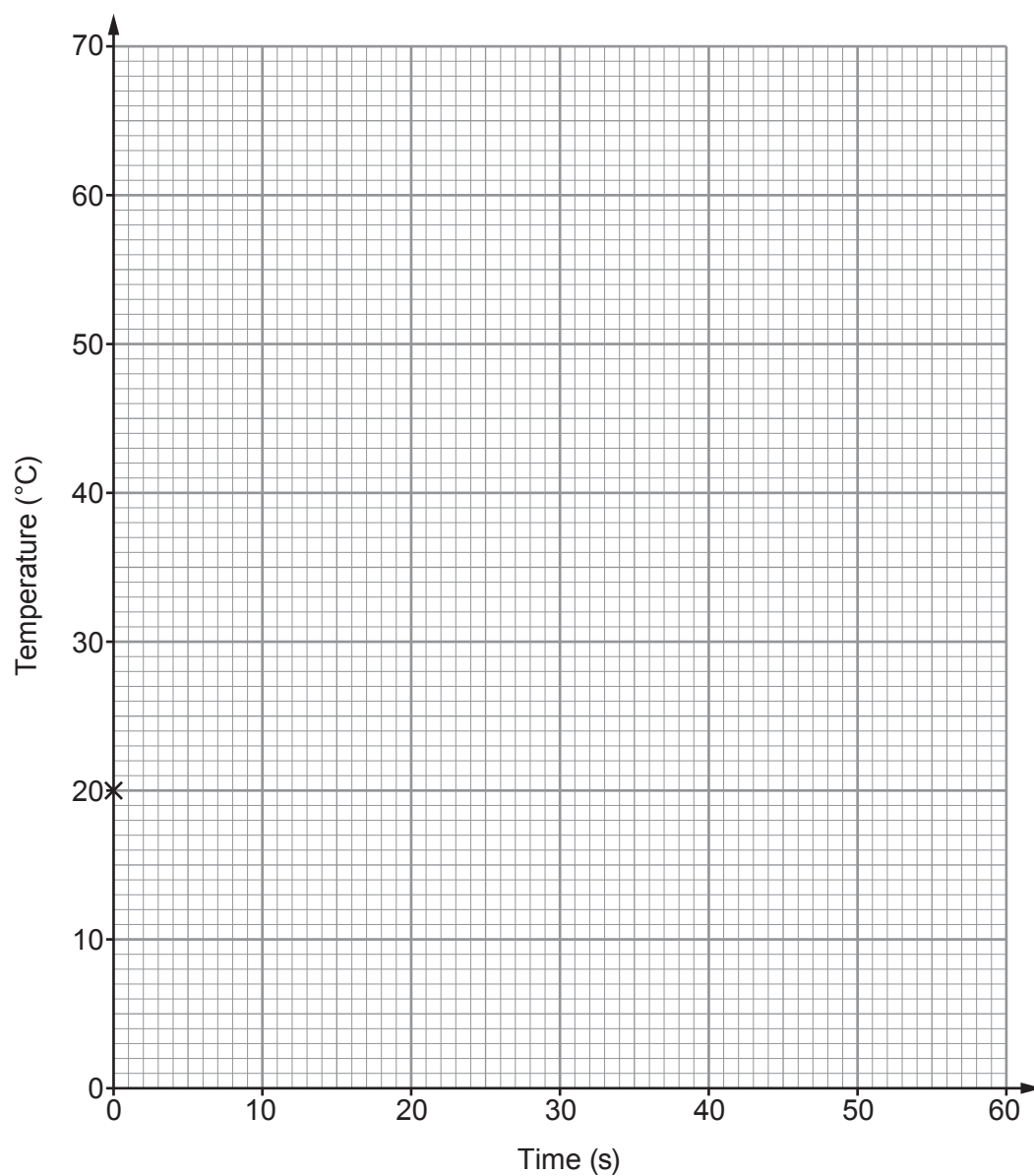


The temperatures are shown in the table below.

Time (s)	Temperature ($^{\circ}\text{C}$)
0	20
10	40
20	50
30	60
40	60
50	58



- (i) Use the data in the table to plot a graph on the grid below and join the points with a ruler. The first point has been plotted for you. [3]



- (ii) Calculate the temperature change during the first 20 seconds. [2]

temperature change = °C



(iii) Complete the following table to describe how the temperature changes over the 50 seconds. The first two rows have been completed for you. [3]

Time (s)	Temperature decreases	Temperature stays the same	Temperature increases
0–10			✓
10–20			✓
20–30			
30–40			
40–50			

(c) Sulfuric acid has the formula H_2SO_4 .

Complete the following table to calculate the relative formula mass of H_2SO_4 . [3]

Element	Number of atoms	Relative atomic mass (A_r)	Total mass of the element in H_2SO_4
hydrogen	2	1	
sulfur	1	32	
oxygen	4	16	64
relative formula mass (M_r)			

